PRACTICAL - 9

Implementation of Prim's Algorithm

Aim of the Practical:

To implement Prim's algorithm to find a Minimum Spanning Tree (MST) of a connected weighted undirected graph and analyze its complexity.

Program:

```
#include <iostream>
#include <vector>
#include <climits>
using namespace std;
int main() {
    const int V = 5;
    int g[V][V] = {
        {0,2,0,6,0},
        {2,0,3,8,5},
        {0,3,0,0,7},
        {6,8,0,0,9},
        {0,5,7,9,0}
    };
    vector<int> key(V, INT_MAX), parent(V, -1);
    vector<bool> used(V, false);
    key[0] = 0;
    for (int count = 0; count < V - 1; count++) {
        int u = -1;
        for (int v = 0; v < V; v++)
            if (!used[v] && (u == -1 || key[v] < key[u])) u = v;
        used[u] = true;
        for (int v = 0; v < V; v++)
            if (g[u][v] && !used[v] && g[u][v] < key[v]) {
                parent[v] = u;
                key[v] = g[u][v];
            }
    }
    int cost = 0;
    for (int v = 1; v < V; v++) {
        cout << parent[v] << " - " << v << " : " << g[v][parent[v]] << endl;
        cost += g[v][parent[v]];
    }
    cout << "MST cost = " << cost;
}
```

Output:

```
0 - 1 : 2
1 - 2 : 3
0 - 3 : 6
1 - 4 : 5
MST cost = 16
```

Time Analysis:

Step-by-Step Derivation:

Let V be the number of vertices. This implementation uses an adjacency matrix and a linear scan to select the minimum-key unvisited vertex.

The outer loop executes V - 1 times.

During each iteration, selecting the next minimum-key vertex scans all V vertices.

$$\text{Selection cost} = (V - 1) \cdot \Theta(V) = \Theta(V^2)$$

After selecting u, the algorithm scans row u of the adjacency matrix, again checking V possible neighbors.

$$\text{Relaxation cost} = (V - 1) \cdot \Theta(V) = \Theta(V^2)$$

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$$T(V) = \Theta(V^2) + \Theta(V^2) = \Theta(V^2)$$

The matrix itself requires $\Theta(V^2)$ storage; the additional key, parent and used arrays require $\Theta(V)$.

Final Time Analysis:

- Time Complexity (matrix implementation): $\Theta(V^2)$.
- Auxiliary Space excluding graph: $\Theta(V)$.
- Graph storage using matrix: $\Theta(V^2)$.

Result:

Prim's algorithm produced an MST of total cost 16 for the sample graph.